the pressure. By “equation of state” is meant the relationship between the resistance and the density, that is, the resistance *as a function* of density.)

In late 1934, when Chandrasekhar took up Ambartsumian’s challenge, the equation of state for white-dwarf matter was known, thanks to calculations by Edmund Stoner of Leeds University in England and Wilhelm Anderson of Tartu University in Estonia. The Stoner–Anderson equation of state showed that, as the density of the white-dwarf matter is squeezed higher and higher, moving from the nonrelativistic regime of low densities and low electron speeds into the relativistic domain of extremely high densities and electron speeds near the speed of light, the matter’s resistance to compression decreases smoothly from 5/3 to 4/3 (left half of Figure 4.3). The resistance could not have behaved more simply.

To meet Ambartsumian’s challenge, Chandrasekhar had to combine this equation of state (this dependence of resistance on density) with the star’s law of balance between gravity and pressure, and thereby obtain a *differential equation*[5] describing the star’s internal structure—that is, describing the variation of its density with distance from the star’s center. He then had to solve that differential equation for a dozen or so stars that have central densities spanning the range from low to extremely high. Only by solving the differential equation for each star could he learn the star’s mass, and see whether it is less than 1.4 Suns.

